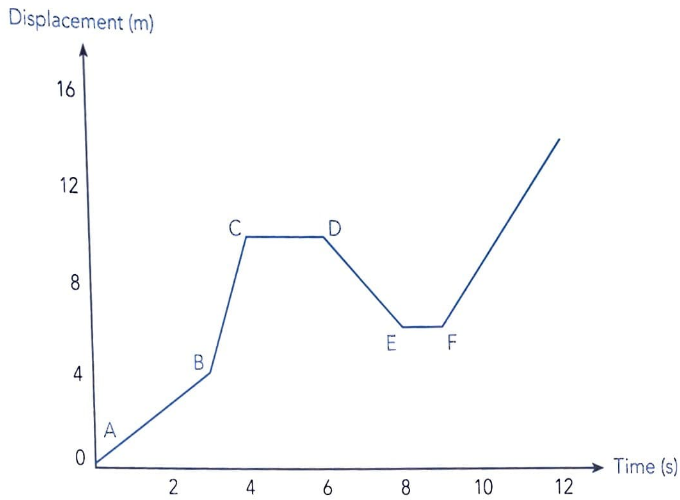
TRIAL TEST 4:
LINEAR MOTION AND FORCE

Time allowed: 60 minutes		**Section One** – Short response	18 marks
Total marks: 60		**Section Two** – Problem solving	30 marks
		Section Three – Comprehension	12 marks

SECTION ONE – SHORT RESPONSE (18 MARKS)

1. Consider the following *s* / *t* graph and answer the questions below.



(a) Describe the state of motion represented by the intervals:

(i) A → B ______________________

(ii) C → D ______________________

[1 mark]

(b) Determine the velocity during the intervals:

(i) A → B ______________________

(ii) E → F ______________________

[1 mark]

(c) Determine the average velocity for the whole journey

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[2 marks]

2. List the following as either vector or scalar quantities.

Speed, mass, acceleration, weight, area, energy temperature, velocity.

Vectors____

Scalars ____ [2 marks]

3. Michael and Aaron are pulling on a cart with forces of 80.0 N West and 60.0 N East. Draw a vector diagram for this situation and find the resultant force.

[3 marks]

4. A skydiver dives from an aircraft in flight and initially goes into free fall without opening her parachute. After achieving maximum velocity she opens the parachute as she prepares to reach the ground safely.

Draw force diagrams to show the forces acting on the skydiver in each of the following cases. Also in each case indicate the resultant force.

(a) Freefall – just jumped

(b) Freefall – at terminal velocity

(c) Parachute just opened

[3 marks]

5. Jeremy throws a ball vertically into the air and it reaches a maximum height of 25.0 m above the point that he released it.

(a) Calculate the initial velocity of the ball.

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[1 mark]

(b) As the ball returns Jeremy allows it to fall to the ground rather than catching it. If the velocity of the ball as it reaches the ground is $22.8\ ms^{-1}$ determine the height from which Jeremy initially threw the ball.

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[3 marks]

6. Paula averaged a power output of 425 W when she won a race to run up the 24 flights of stairs in a building. If each flight of stairs is separated by 3.00 m and Paula has a mass of 55.0 kg, calculate the time taken for her to reach the top of the stairs.

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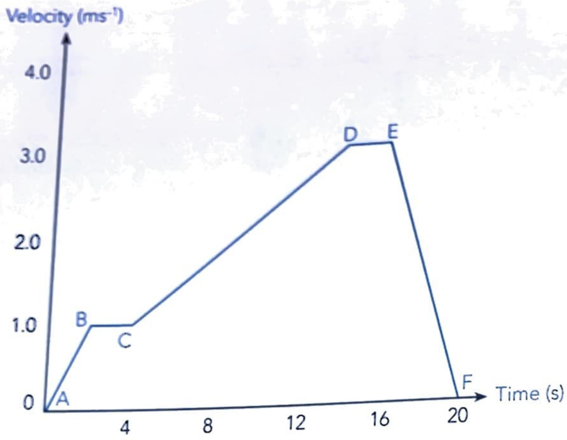
[2 marks]

SECTION TWO – PROBLEM SOLVING (30 MARKS)

7. **Consider the following *v/t* graph and answer the questions below.**



(a) Describe the state of motion represented by the intervals:

(i) A → B ______________________________

(ii) E → F ______________________________

[2 marks]

(b) Determine the displacement:

(i) at $t = 4.0$ s.

____

____

(ii) during the fourth second.

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[3 marks]

(c) (i) When is acceleration a maximum?

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(ii) Determine acceleration at $t = 10.0$ s.

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____

[3 marks]

8. John is going for his favourite bike ride at a steady 18.0 kmh^{-1} when he passes his stationary friend Jason. Jason immediately sets off after John and accelerates at 2.4 ms^{-2} until he reaches his top speed 3.0 s later. He then continues at this top speed as he cycles towards John.

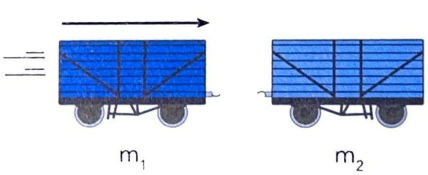
(a) Does Jason catch John? Show clearly all calculations.

[5 marks]

(b) If so, how far did he travel before he reached John?

[5 marks]

9. In the railway marshalling yards a carriage of mass 5.0×10^4 kg, m_1 , was shunted against a stationary smaller carriage, m_2 , which has half the mass of the shunted carriage. The heavier carriage was travelling at 4.0 ms^{-1} and the two couple together on impact.



(a) Calculate the velocity with which the two carriages move off;

[4 marks]

(b) Compare the kinetic energy of the larger carriage before the coupling with that of both carriages after coupling.

[4 marks]

(c) Has the principle of conservation of energy been obeyed? Explain your answer.

[4 marks]

SECTION THREE – COMPREHENSION (12 MARKS)

10. During a physics experiment to obtain data for a velocity-time graph for Bronwyn riding her bicycle on a level road, Mike wrote the following:

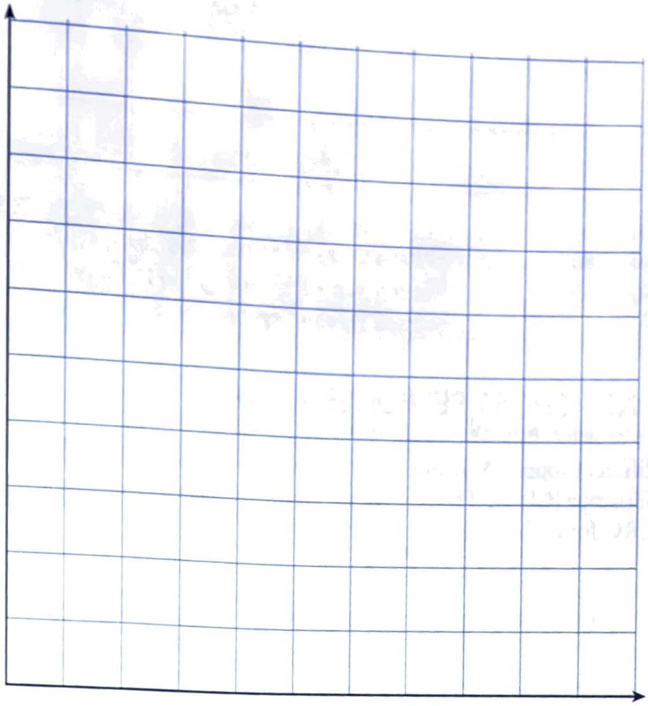
Bronwyn accelerated her bike uniformly from rest and called out when she reached a velocity of 6 ms⁻¹. This took her 10 s. For the next 10 s she maintained a constant velocity of 6 ms⁻¹ before accelerating uniformly for 5 s when she told me afterwards that her velocity was 14 ms⁻¹. She maintained this velocity for a further 15 s before applying the brakes uniformly. It took her 10 s to stop.

(a) Use what Mike wrote to:

(i) tabulate the data

[2 marks]

(ii) draw a velocity-time graph.



[3 marks]

(b) Use your graph to determine the:

(i) acceleration between 20 s and 25 s

[2 marks]

(ii) deceleration when braking

[2 marks]

(iii) total distance travelled by Bronwyn.

[3 marks]

END OF TEST (60 MARKS)